

Cyclotomic Polynomial

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The n th cyclotomic polynomial, $P_n(X)$ is the polynomial in $C[X]$ (complex numbers) which is the product of all $(x - r)$, over r a primitive n th root of unity - ie. an element of μ_n (the n th roots of 1, thought of inside the complex numbers) whose order in this finite group is n .

Problem 1. Explain why $P_n(X)$ divides $X^n - 1$ in $C[X]$. Explain why $X^n - 1$ has no multiple roots.

Proof. All the roots of $P_n(X)$ are by definition primitive n th roots of unity thus they are n th roots of unity. All n -th roots of unity are roots of $x^n - 1$ thus all the roots of P_n are roots of $x^n - 1$, thus P_n must divide $x^n - 1$.

Assume the contrary, $f(x) = x^n - 1$ has a repeated root. Thus by definition of a repeated root, $f(r) = 0$ and $f'(r) = 0$. Thus $r^n - 1 = 0$ and $nr^{n-1} = 0$, since $\text{char } \mathbb{C} = 0$, since $n \neq 0$, the derivative isn't trivially zero. $nr^{n-1} = 0 \implies r = 0$, but 0 is not a root of $x^n - 1$, $0^n - 1 = -1$, which is a contradiction. Thus by contradiction $x^n - 1$ has no repeated roots. $\square$

Problem 2. Suppose p is prime. Explain why $P_p(X) = (X^p - 1)/(X - 1) = 1 + X + X^2 + \dots + X^{p-1}$.

Proof. Suppose p is prime. All the natural numbers n which are relatively prime to p are all natural numbers less than p . Let $\omega = e^{2\pi i/n}$. Thus $\forall j \in \mathbb{N} \ni j < p$, $\text{ord}(\omega^j) = n$. Thus $(x - \omega^j)$ is a factor to $P_p(X)$. Thus $P_p(x) = \prod_{j=1}^{p-1} (x - \omega^j)$. Since $P_p(x) \mid x^p - 1$ and since it has $p - 1$ of the linear factors of $x^p - 1$ and the only other linear factor is $x - 1$, since $x^p - 1 = \prod_{n=0}^{p-1} (x - \omega^n)$. $P_p(X) = \frac{x^p - 1}{x - 1}$. After doing the long division we see, $P_p(x) = 1 + x + x^2 + \dots + x^{p-1}$. $\square$

Problem 3. Prove $P_n(X)$ is a monic integer polynomial. Hint: Induct on n . If $n=1$ its obvious. Otherwise Let $H(X)$ be the product of all the P_k for $k \mid n$ and $k < n$. H is monic integer by induction. Explain why H divides $X^n - 1$ in $\mathbb{C}[X]$, in $\mathbb{Q}[X]$, finally, in $\mathbb{Z}[X]$. Explain why $X^n - 1 = P_n H$, and conclude that P_n is integer monic.

Proof. Let $\omega_n := e^{2\pi i/n}$.

Base Case: $P_1(X) = x - 1$.

Induction Hypothesis: Suppose $\forall k \in \mathbb{N} \ni k < n$, P_k is a integer monic polynomial.

Inductive Step: Let $H(x) = \prod_{j < n, j \mid n} P_j(x)$. By induction, $\forall j \in \mathbb{N} \ni j \mid n \implies j < n$, thus P_j is integer monic. Thus $H(x)$ is also integer monic.

To show that $H(x) \mid x^n - 1$, it suffices to show that $\forall j \in \mathbb{N} \ni j < n$ and $j \mid n$, $P_j(X) \mid x^n - 1$. Since $j \mid n$, $\omega_j = e^{2\pi i/j} = e^{2\pi i n/nj} = e^{2\pi i(n/j)/n} = \omega_n^{n/j}$. Moreover since $j \mid n$, $n/j \in \mathbb{Z}$ and $n/j \mid n$. Thus $x - \omega_j$

is a factor of $x^n - 1$ and thus $\forall a \in \mathbb{N} \ni a < j$, ω_j^a is also a root to $x^n - 1$. Thus $P_j(X) | x^n - 1$ in $\mathbb{C}[X]$ and thus $H(x) | x^n - 1$.

Since $H(x)$ and $x^n - 1$ are integer polynomials, $\frac{x^n - 1}{H(x)} \in Q[X]$ and $H(x) | x^n - 1$ in $Q[X]$. Since $H(x)$ is integer monic, it is primitive as $\gcd$ is automatically 1 because of the leading term. By Gauss's Lemma, $H(x) | x^n - 1$ in $Z[x]$.

Finally, $\forall j \in \mathbb{N} \ni j < n$, ω^j is a root of $x^n - 1$. Furthermore either j and n are relatively prime or they are not. If $(j, n) = 1$, by definition, ω^j is a root of P_n . Assume the contrary, ω^j is a root of P_a for some $a < n \ni a | n$. All roots of P_a are ω_a^b where $\gcd(b, a) = 1$. However since $\omega_a^b = e^{2\pi i b/a} = e^{2\pi i b \frac{n}{a}/n}$. Thus $j = b \frac{n}{a}$, but since $n/a | n \implies \gcd(j, n) = \frac{n}{a}$, which is a contradiction.

If $(j, n) \neq 1$, let $k = (j, n)$. Since $k | n \implies k' = \frac{n}{k} | n$. $\omega_n^j = e^{2\pi i j/n} = e^{2\pi i \frac{j}{k}/k'} = \omega_{k'}^{\frac{j}{k}}$. Thus ω_n^j is a root of $P_{k'}$ which is a factor of H .

Thus all the linear factors of $x^n - 1$ are either in P_n or H . Thus $x^n - 1 = P_n(x)H(x)$. Thus $P_n(x)$ is integer monic.

Thus by induction, $\forall n \in \mathbb{N}$, $P_n(x)$ is integer monic. □

Claim 4a. *Raising to the power mod p is a ring homomorphism.*

Proof. $\forall g, h \in F_p[X]$,

$$\begin{aligned} (g(x) + h(x))^p \mod p &= \sum_{n=0}^p \binom{p}{n} g(x)^n h(x)^{p-n} \mod p \\ &= g(x)^p + \sum_{n=1}^{p-1} \frac{p!}{n!(p-n)!} g(x)^n h(x)^{p-n} + h(x)^p \mod p \\ &= (g(x)^p \mod p) + (h(x)^p \mod p) \end{aligned}$$

and,

$$(g(x)h(x))^p \mod p = g(x)^p h(x)^p \mod p$$

Thus raising to the power mod p is a ring homomorphism. □

Problem 4. *Prove P_n is irreducible in $Z[X]$, or equivalently (by Gauss's lemma) in $Q[X]$*

Proof. Suppose f is integer monic polynomial which divides P_n and $\exists h \in Z[x]$, such that $P_n(x) = f(x)h(x)$. Let r be a root of $f(x)$. $f(x) | P_n$ and r is a root of $f(x)$ it is also a root of $P_n(x)$ and thus a primitive root of unity and can generate all other roots of $x^n - 1$ through exponentiation and hence can generate all roots of $P_n(x)$ through exponentiation. Let p be a prime which doesn't divide n .

By the previous homework, since r is a root of $f(x)$ and $f(x)$ is integer monic its maximal polynomial is integer monic. Let the maximal polynomial of r be $g(x)$. Thus $Z[r] \cong Z[x]/(p(x))$. Let $\bar{p}(x) = p(x) \mod p$. By the previous homework, there exists a prime ideal, $P = (\bar{p}(x))$ where $P \cap \mathbb{Z} = (p)$. Let $F = Z[r]/P$, by the previous homework, $F \cong \mathbb{F}_p[x]/(\bar{p}(x))$. By the previous homework, $\bar{p}(x)$ is maximal and thus F is a finite field.

Since $p \nmid n$, nx^{n-1} is a nonzero polynomial. Assume the contrary, nx^{n-1} and $x^n - 1$ are not relatively prime, thus they share a common factor. Thus they must also share a root. Let r be the

common root to both polynomials. $r \equiv 0 \pmod p$ and $r \equiv 1 \pmod p$, which is a contradiction. Thus by contradiction $(nx^{n-1}, x^n - 1) = 1$. Thus $x^n - 1$ has no double roots in $\mathbb{F}_p$ and thus $P_n \pmod p$ has no double roots either, thus each root is either in $h(x) \pmod p$ or $f(x) \pmod p$ and thus $\gcd(h(x) \pmod p, f(x) \pmod p) = 1$, because otherwise they would share a factor and thus a common root which would lead to a contradiction.

Since r is a root of $P_n(x)$, $r = \omega_n^j$ where $(j, n) = 1$. $r^p = \omega_n^{jp}$, since $p \nmid n$, and p is prime, $(p, n) = 1$, thus $(jp, n) = 1$. Thus $r^p = \omega_n^{jp}$ is another root of $P_n(x)$. By claim 4a, $h(x)^p \pmod p = h(x^p) \pmod p$. Thus $h(r)^p \pmod p = h(r^p) \pmod p$. Since h and f are relatively and thus share no common roots, r is not a root of h and thus $h(r)$ is nonzero thus $h(r)^p$ is nonzero and thus $h(r^p)$ is nonzero. Thus r^p is not a root of h . Since h is nonzero over F , it is nonzero over $\mathbb{C}$ because we went down a ring homomorphism from $\mathbb{C}[x] \rightarrow F$. Thus r^p must be a root of f .

Since r is a generic root of f and p is any prime which doesn't divide n , we have shown that for any r , a root of f , and any p , a prime which doesn't divide n , r^p is a root of f . Since r is a root of P_n , every root of $x^n - 1$ is r^a for some $a \in [0, \dots, n-1]$. For r^a to be a root of P_n , $(a, n) = 1$. Since $\mathbb{Z}$ is a ufd, $a = p_1^{k_1} \dots p_m^{k_m}$ where $p_1, \dots, p_m \nmid n$.

By the above, r^{p_1} is a root of $f \implies (r^{p_1})^{p_1}$ is a root of f and so on and so forth and which then implies $r^{p_1^{k_1}}$ is a root of f . This then implies, $(r^{p_1^{k_1}})^{p_2}$ is a root of f which then eventually implies, $r^{p_1^{k_1} p_2^{k_2}}$ is a root of f , etc. Thus r^a is a root of f . Thus every root of P_n is a root of f but not h (f and h are relatively prime and thus share no common roots). Thus $f(x) = P_n(x)$ and $h(x)$ is a unit. and thus $P_n(x)$ is irreducible. $\square$